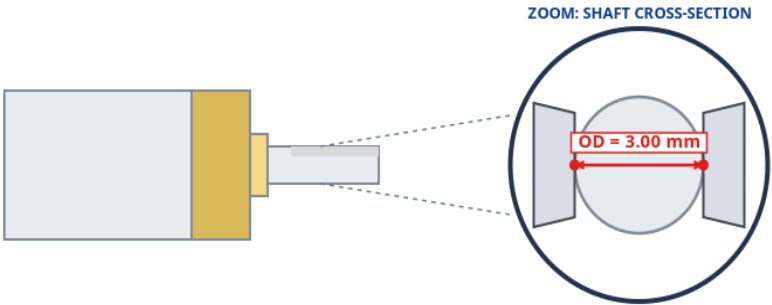


TABLE ROW 1 N20 Shaft Outer Diameter (OD)

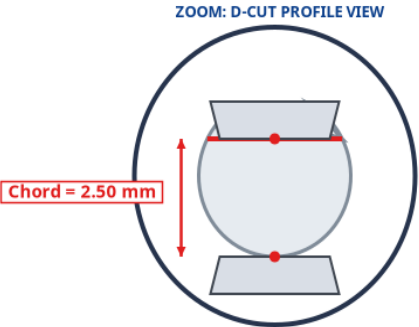
EXPECTED NOMINAL: 3.00 mm (Typical: 2.95 – 2.98 mm) TOOLING: Outside Measuring Jaws (Flat Anvils)



- WHERE TO MEASURE: Clamp jaws across the full cylindrical shaft section near the tip.
- CRITICAL CHECK: Do not measure across the D-flat! Rotor arm bore relies on this exact diameter.

TABLE ROW 2 N20 D-Flat Chord Thickness

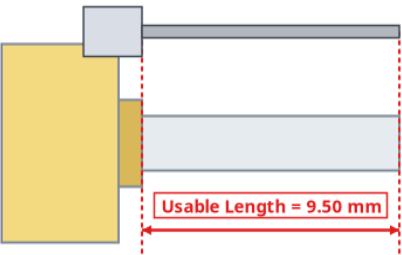
EXPECTED NOMINAL: 2.50 mm (Typical: 2.45 – 2.48 mm) TOOLING: Outside Measuring Jaws (Flat to Curve)



- HOW TO POSITION CALIPER:**
1. Rest the fixed flat anvil of caliper directly on the flat face.
 2. Ensure the jaw sits 100% flush (do NOT tilt or angle).
 3. Gently close moving jaw until it touches the curved back.
 4. Nominal reading is exactly 2.50 mm.
- > If measured < 2.44 mm, rotor arm requires smaller CAD flat.
-> If measured > 2.52 mm, rotor arm flat bore must be enlarged.

TABLE ROW 3 N20 Usable Shaft Length

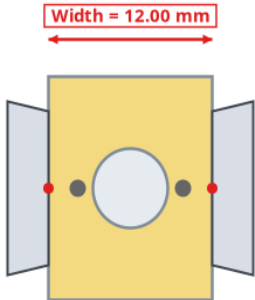
EXPECTED NOMINAL: 9.50 mm (Typical: 9.0 – 10.0 mm) TOOLING: Vernier Depth Probe or Stepped Jaws



- REFERENCE DATUM: Measure from the front face of the brass collar (where rotor hub stops).
- ROTOR LIMIT: The 3D printed rotor arm hub must not exceed this length to prevent shaft end rubbing.

TABLE ROW 4 N20 Motor Body Width

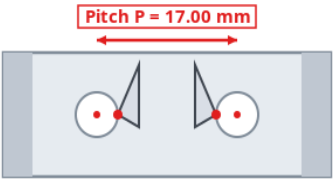
EXPECTED NOMINAL: 12.00 mm (Typical: 11.9 – 12.1 mm) TOOLING: Outside Measuring Jaws (Across Gearbox Flange)



- CRITICAL FIT IMPORTANCE:**
- Metal U-Bracket Clearance: The stamped metal bracket cradle has an internal gap of ~12.2 mm.
 - If the motor body is > 12.1 mm, it will bind inside the bracket.
 - If the motor body is < 11.9 mm, use thin shim tape to prevent radial rocking under 600 RPM eccentric unbalance.

TABLE ROW 5 N20 Bracket Base Hole Pitch

EXPECTED NOMINAL: 17.00 mm (Pitch Between Base Slots) TOOLING: Inside Nibs (Tangent Method: L_outer - D)



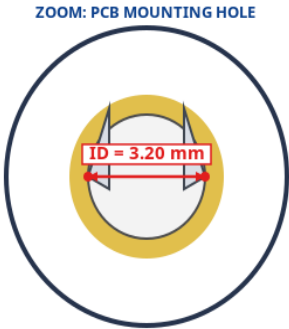
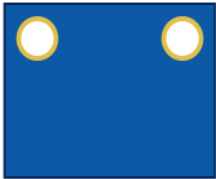
EXACT CALIPER PITCH FORMULA:

- Method 1: Pitch = L_inner + D_hole
- Method 2: Pitch = L_outer - D_hole

-> Guarantees ±0.02 mm accuracy without guessing centerlines!

TABLE ROW 6 ADXL345 PCB Hole Diameter

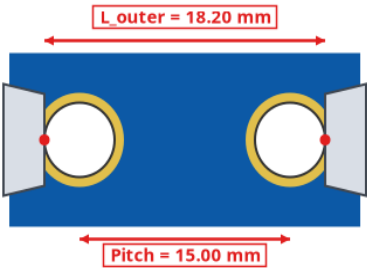
EXPECTED NOMINAL: 3.20 mm (M3 Clearance Check) TOOLING: Inside Measuring Nibs (Delicate Insert)



- PASS CRITERION: M3 screw shank (OD 2.92 mm) must pass through without catching.
- CAUTION: Do not force inside nibs with high pressure; PCB annular copper ring can deform.

TABLE ROW 7 ADXL345 Hole Center Pitch

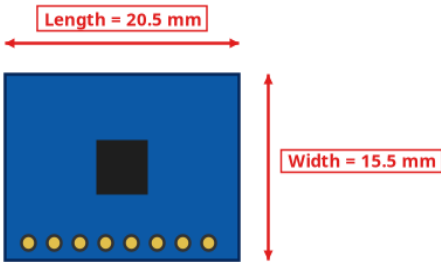
EXPECTED NOMINAL: 15.00 mm (Dictates Sensor Bracket Spacing) TOOLING: Inside Nibs or Outside Tangent Jaws



- VERIFICATION CALCULATION:**
1. Measure outside span: L_outer = 18.20 mm
 2. Measure single hole ID: D = 3.20 mm
 3. Compute True Pitch: P = 18.20 - 3.20 = 15.00 mm
- > CRITICAL CHECK: Sensor bracket upright holes must be designed at exactly this measured pitch (15.00 mm) ±0.05 mm.

TABLE ROW 8 ADXL345 PCB Dimensions (W × L)

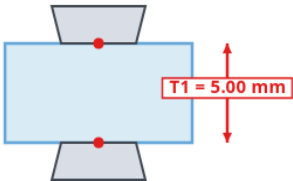
EXPECTED NOMINAL: 15.5 mm × 20.5 mm (Thickness: 1.6 mm) TOOLING: Outside Measuring Jaws (Substrate Outline)



- SIDE FLANGE & HEADER CLEARANCE:**
- PCB Substrate Thickness: 1.60 mm.
 - Rear Solder Joints: Pins protrude 1.0 – 1.5 mm on back.
 - 3D Bracket Recess: The vertical mount wall **MUST** have a recessed pocket or standoffs to prevent the solder blobs from preventing flush bracket contact.

TABLE ROWS 9 & 10 Acrylic Sheet 1 & 2 Individual Thickness

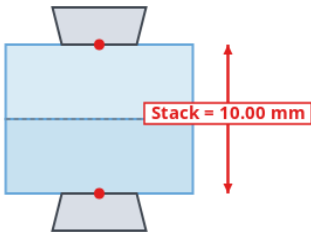
EXPECTED NOMINAL: 5.00 mm per sheet (Expected: 4.6 - 5.2 mm) TOOLING: Outside Measuring Jaws (Check All 4 Corners)



- 4-CORNER THICKNESS PROTOCOL:**
- 1. Measure Sheet 1 at Corner 1, 2, 3, 4.
 - 2. Record average in Trial 1 column.
 - 3. Repeat independently for Sheet 2 in Trial 2 column.
- Cast PMMA Tolerance: Cast acrylic has ±10% thickness variation.
 - A sheet labeled '5 mm' often measures 4.75 mm or 5.15 mm.
 - > Knowing true gauge prevents bolt thread bottoming!

TABLE ROW 11 Combined Stack Thickness (Dual 5 mm PMMA)

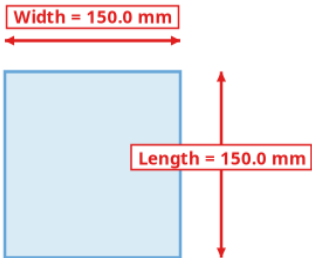
EXPECTED NOMINAL: 10.00 mm Total (Determines M3 Bolt Grip) TOOLING: Outside Measuring Jaws (Clamped Sandwich)



- FASTENER ENGAGEMENT CALCULATION:**
- Clamped Stack Thickness = 10.00 mm
 - Bracket Foot Flange = 3.50 mm
 - DIN 125 Washer = 0.50 mm
 - M3 Nylon Locknut Height = 4.00 mm
- > TOTAL GRIP LENGTH REQUIRED = 18.0 mm
- > VERIFICATION: Standard M3 x 20 mm bolts provide exactly 2 mm thread protrusion past the nylon ring (SAFE)!

TABLE ROWS 12 & 13 Acrylic Blank 1 & 2 Width × Length

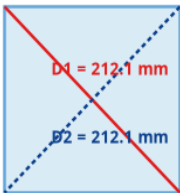
EXPECTED NOMINAL: 150.0 mm × 150.0 mm (Square Cut Check) TOOLING: Caliper Long Beam (Check Edge Flushness)



- EDGE FLUSHNESS CHECK:**
- 1. Stack Blank 1 on Blank 2.
 - 2. Run finger along all 4 perimeter seams.
 - 3. Measure edge overhang (step mismatch).
- PASS CRITERION: Overhang must be < 0.5 mm.
 - If edges differ, file the wider edge flush BEFORE drilling to ensure 1:1 sticker template lines up on both plates.

TABLE ROW 14 Acrylic Squareness Diagonals (D1 vs D2)

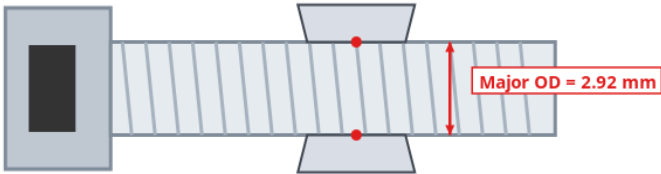
EXPECTED NOMINAL: 212.13 mm (sqrt(150² + 150²)) TOOLING: Caliper Diagonal Span



- SQUARENESS VERIFICATION RULE:**
- Pythagorean Theorem: sqrt(150² + 150²) = 212.13 mm.
 - If D1 == D2 (within ±0.5 mm): The blank is a TRUE SQUARE.
 - If D1 != D2 (e.g. D1=213.5, D2=210.7): The blank has PARALLELOGRAM SKEW.
- > If skewed, do NOT use outside edges for hole reference; use the taped 1:1 sticker template crosshairs exclusively!

TABLE ROW 15 M3 Bolt Thread Shank Major Diameter

EXPECTED NOMINAL: 2.92 mm (Commercial Range: 2.87 – 2.98 mm) TOOLING: Outside Measuring Jaws (Across Thread Crests)

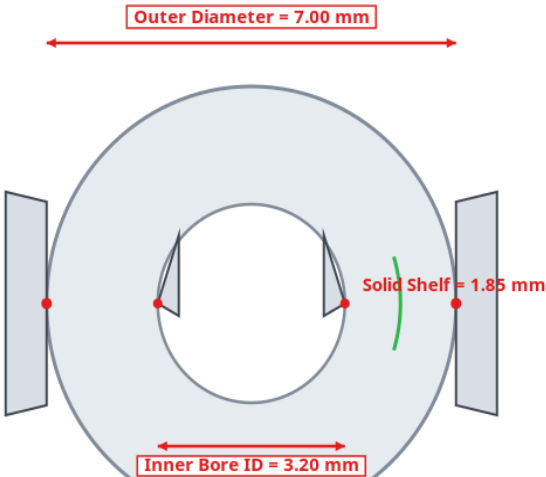


CRITICAL CLEARANCE PHYSICS (M3 IN Ø3.3 mm HOLE):

- 1. Free Slip Verification: Thread major OD must measure < 2.98 mm.
In our Ø3.30 mm drilled hole, this yields exactly 175 µm radial clearance.
- 2. Zero Binding: The bolt will slide through both 5 mm acrylic sheets without requiring any clockwise twisting or forcing.
- 3. Brittle Failure Prevention: If an M3 screw is forced into a Ø3.0 mm hole, the thread crests bite and exert hoop tensile stress > 70 MPa, cracking acrylic!
-> RESULT: Ø3.3 mm hole + 2.92 mm bolt = 100% reliable, crack-free mounting.

TABLE ROW 16 DIN 125 M3 Flat Washer Dimensions

EXPECTED NOMINAL: Outer Diameter: 7.00 mm • Bore ID: 3.20 mm TOOLING: Outside Jaws (OD) & Inside Nibs (ID)



BEARING SURFACE AREA & PULL-THROUGH SAFETY:

- 1. Bearing Shelf per Side = (OD - Hole_Dia) / 2 = (7.00 - 3.30) / 2 = 1.85 mm.
- 2. Effective Bearing Area = $\pi/4 * (7.0^2 - 3.3^2) = 29.93 \text{ mm}^2$.
- 3. Contact Retention: Retains 98.4% of standard ISO bearing area.
- 4. Clamping Force Distribution: 4 tightened corner M3 washers distribute
~3,500 N force uniformly across 120 mm² of acrylic, preventing crazing.
-> ZERO RISK OF BOLT PULL-THROUGH even under intense motor vibration!